PRECISION TIMING / ENGINEERING SOFTWARE

# MHM2010 / T4
# Maser Timing Workbench

## Operator and Scientific Manual

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FITS and Franks timing, reference 1 PPS figures, MHM2010 DDS/drift and

T4 iMaser 3000 SETFSREL/DRIFTCOMP suggestions, T4 MONIT telemetry, stability and native Veusz output against MJD and local ISO time.

| | |
|---|---|
| Application version | 0.2.0 (0.1.0 content retained below) |
| Bundled writer | `vszh5_writer.py` 0.0.24 (built on master 0.0.23) |
| Author | William W. Wallace |
| Date | November 3, 2026 |
| Runtime | Python 3.12 |
| GUI | qtpy / PySide6 / pyqtgraph |
| License | GPL-3.0-or-later |

**Review boundary.** This application never sends a setting to a maser. The T4 panel can send only the documented read-only queries MONIT, RDFS, GETID and RDREG. Numerical candidates are advisory and require verification of the current settings, sign conventions, reference behavior and application epoch. Historical examples are not settings to enter today.

Deliverable contract: runnable source; live interval fitting; both drift units; FITS and Franks timing inputs; HMD context; stability; Stable32; native `.vszh5`; bookmarked PDF and editable LuaLaTeX; tested examples and a 7z package. Physical instrument certification is outside this release.

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## Purpose and verified instrument relationship

The instrument mapping follows the MHM2010 manufacturer's guide [1]; stability definitions follow NIST SP 1065 [2]; and interchange files follow the Stable32 manual [3]. These references establish the analytical framework, not a certification of the installed system.

The workbench estimates residual maser-to-reference fractional frequency and frequency drift from measured timing offsets. Housekeeping telemetry identifies instrument context and changes; it is not an independent frequency reference. The Green Bank system logs switched one-pulse-per-second comparisons involving Site1Hz, MicroSem and GPS channels [4].

The MHM2010 manual gives the exact relationship between maser oscillation frequency, the external synthesizer number and the standard output frequency. An increase in the synthesizer number lowers output frequency; the nominal calibration is $7.04024 \times 10^{-17}$ per least-significant displayed digit [1]. The application derives its local gain from the exact denominator, rather than treating the nominal calibration as exact.

The supplied screenshot identifies the two target controls as **External DDS Setting** and **Drift Correction**. The manufacturer's manual explicitly offers **DDS Units/day** and **pp** $10^{18}$**/day** in the drift control [1]. Both are computed simultaneously in this release.

| Documented item | Treatment in this release |
|---|---|
| DDS versus output direction | Increasing DDS decreases output frequency [1]. |
| DDS precision | One displayed last digit is $10^{-7}$ in the synthesizer number [1]. |
| Drift-unit choices | Both DDS units/day and parts in $10^{18}$/day are available [1]. |
| External control | Front-panel synthesizer selection must permit external control [1]. |
| Drift-field sign | Not explicitly stated in inspected Rev. J; user selects and verifies it. |
| Numeric limits and cadence | Not specified in inspected Rev. J; not invented by this program. |
| Housekeeping selector codes | Retained as raw logged values; not decoded into an enable assertion. |

**Review boundary.** A phase offset cannot be removed merely by selecting a zero steady frequency error. This software does not plan phase slews or write the instrument. It estimates slope and curvature and leaves the constant phase offset unchanged.

### Release structure

- **Application**: GUI entry point plus separate input, science and export modules.
- **Batch path**: `analyze_maser.py` accepts explicit paths and historical assumptions.
- **Writer**: the supplied native h5py writer, with a targeted date-shape fix.
- **Documentation**: this PDF, its LuaLaTeX source, README, changelog and validation report.
- **Examples**: a 30-day historical result with raw pair datasets, reports and Stable32 output.

## Installation and first run

Use Python 3.12 in an isolated environment. No fixed personal directory names are embedded in the application. Run the following inside the extracted package:

```
py -3.12 -m venv .venv
.\.venv\Scripts\python.exe -m pip install -r requirements.txt
.\.venv\Scripts\python.exe MHM2010_Correction_App.py
```

The supplied Windows launcher uses the adjacent environment if present, otherwise Python's 3.12 launcher. Source files have author/revision headers, semantic versions, Spyder cell markers and explanatory comments. In Spyder, select the same environment; a dedicated console is preferable if a GUI loop is already active.

The default backend is PySide6 through qtpy. The application does not import PySide6 directly. pyqtgraph is configured to match the selected Qt backend. PyQt6 may be selected explicitly through `QT_API`, but that alternate backend was not validated for this release.

### Dependencies and offline behavior

The runtime requirements are NumPy, SciPy, pandas, Astropy, h5py, QtPy, PySide6 and pyqtgraph. The supplied environment-version list records the packages used for testing. Installing dependencies requires package access; the scientific analysis itself requires no web service, API key, instrument socket or serial connection.

Veusz is optional for viewing or rendering output, not for generating `.vszh5`. Test dependencies include pytest and AllanTools. The included manual was compiled with LuaLaTeX; run the command twice from the docs folder:

```
lualatex -interaction=nonstopmode -halt-on-error MHM2010_Workbench_Manual_v0.1.0.tex
```

### Quick workflow

1. Select raw OnePpsDeltas FITS/FITS.GZ files using the GUI or drag-and-drop.
2. Load HMD housekeeping separately; select the intended MicroSem/GPS pair.
3. Choose a continuous acquisition run and a clean fit interval.
4. Compare linear and quadratic behavior and inspect residuals and other references.
5. Enter the DDS and enabled drift at the intended application epoch.
6. Verify signs and operating state; inspect both drift-unit representations.
7. Compute stability with an explicit sampling tolerance and detrending choice.
8. Export to a new results folder and retain the source hashes and review record.

**Review boundary.** This release was tested in a Linux cloud environment. Actual Windows file dialogs, display scaling, Stable32 application import and physical maser operation remain untested. No connected bracca device was exposed at the last inventory check, so local staging was not performed.

# Input data and provenance

## Preferred input: raw OnePpsDeltas FITS

The parser requires the `OnePpsDeltas` binary table with `DMJD`, `CHANNELA`, `CHANNELB` and `DELTAT`. It verifies UTC time metadata and declared time/phase units. Files marked simulated are rejected. Blank channel names, nonpositive timestamps and padding rows are rejected. FITS `TNULL` values are honored when supplied.

Every measured channel pair stays separate. The direct `MicroSem -> CNS_GPS1` pair is preferred over subtracting sequential Site1Hz observations. Reverse-direction pairs are not merged into the direct pair: their measurements occur at different times and can include fixed channel-path asymmetries. Valid raw zero values and $-9999$ ns observations are not automatically treated as missing.

Rows are sorted by timestamp. Identical duplicates are removed and counted; different values at one identical pair timestamp cause a failure rather than an arbitrary choice. The parser records input basenames, sizes and SHA256 hashes, without exporting personal absolute directory paths.

## Alternative input: Franks t00new

Franks files are reduced timing data. The parser locates the named MJD header, requires the nanosecond declaration and preserves each available timing column. Values $-9999$ and the historical $-999$ sentinel are removed in this adapter only. Raw FITS and reduced Franks files cannot be loaded as one combined batch.

Hourly median timestamps are often irregular. They can be fitted using actual timestamps, but cannot automatically be treated as a uniform stability record. The original reduction windows and raw samples cannot be recovered from the reduced values.

## HMD telemetry

HMD is parsed as a timestamped comma-delimited table. It must include MJD and the external synthesizer column. All numeric housekeeping channels remain available for display. Large discrete DDS changes and selector-code changes are marked as candidate events; gradual drift ramps are not called steps.

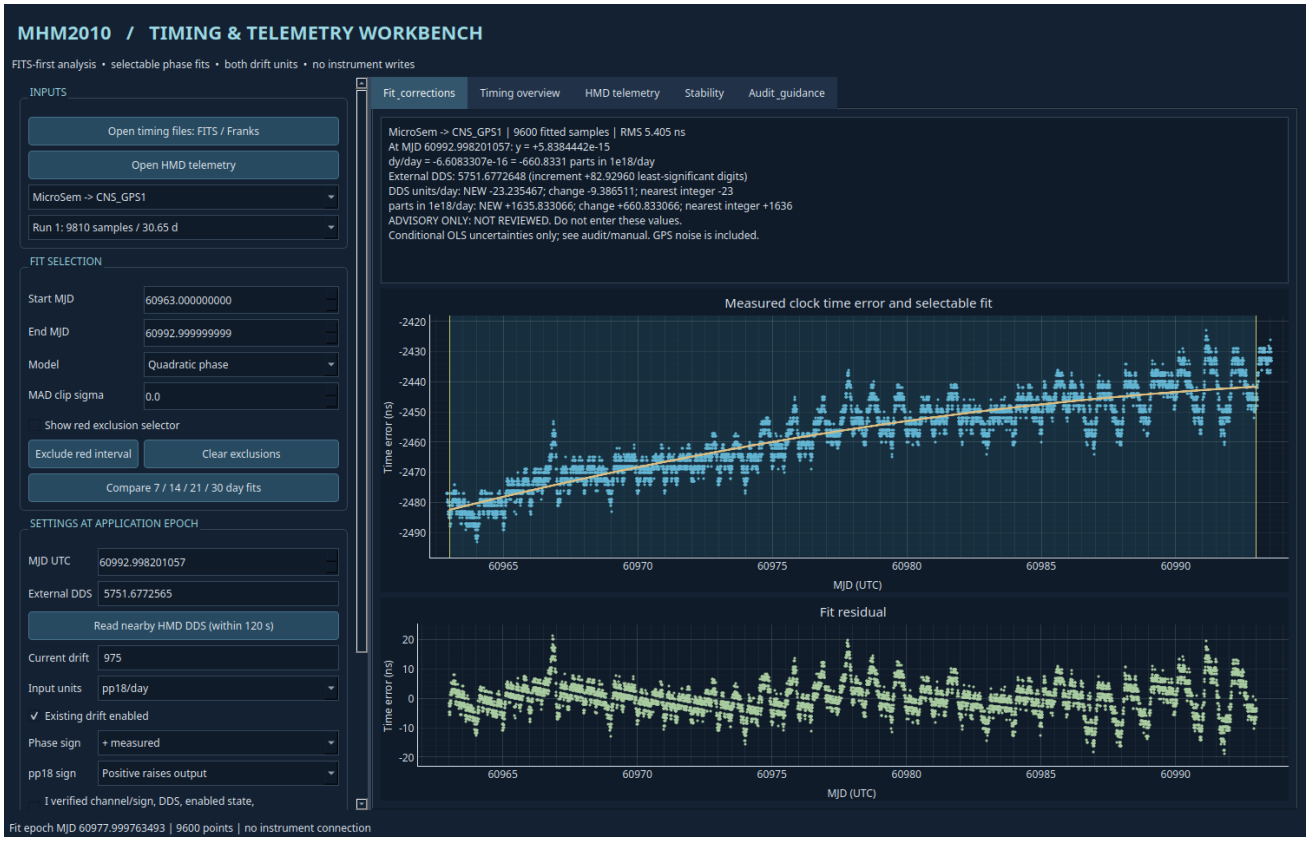
The event detector uses a configurable function argument with a GUI default of 50 least-significant DDS digits. This is an application heuristic, not a manufacturer alarm threshold. Alarm/status channel voltages remain raw values. Missing HMD coverage is reported; historical telemetry is never silently used as the current state at an unrelated epoch.

## Timing overview and the supplied Veusz exemplar

The overview creates UTC-hour medians for available Site1Hz comparisons, direct MicroSem/GPS measurements and a separately labeled difference of Site1Hz medians. All raw measured pairs remain in the export. Missing channels such as RA or TAC are not fabricated.

Each available overview channel has MJD and proper UTC-axis export pages. This reproduces the useful dataset families in the supplied Franks Veusz example without copying its all-missing channels or invalid legacy dates. UTC-hour binning is not claimed identical to the historical reduction windows.

# Interactive analysis




**Fit and corrections tab.** The blue region selects the fitting interval. The phase and residual plots share their horizontal view. Numeric MJD limits provide precise selection without requiring a mouse.

The red region is an independent exclusion selector. Add its interval to the exclusion list or clear the list. Excluded observations remain in the input and export, with an explicit used-in-fit mask. Optional iterative MAD residual clipping is disabled by default. It can reject outliers, but it changes noise statistics and must not conceal physical events.

A linear phase model estimates frequency only; the drift field is not estimated and should not be changed on that basis. A quadratic phase model estimates frequency and frequency drift. Internal scaling improves numerical conditioning without changing the physical units of the returned coefficients.

The 7/14/21/30-day comparison button reports trailing-window fits in the audit tab. These are unclipped quadratic diagnostic fits, with actual span and sample count shown. They are not confidence intervals.

## State and stale-result handling

Changing the fit interval/model invalidates old stability results. Settings can be saved without source-directory paths, then restored after selecting the original files. Restoring settings clears operator acknowledgments. Export creates a new folder; it never silently replaces an earlier result directory.

**Review boundary.** The plots are interactive analysis of selected files, not streaming acquisition. Large file reads run outside the GUI thread; fitting, overview construction and export may briefly occupy the GUI thread for very large datasets. There is no serial or network maser interface in this release.

# Correction interpretation and manual review

## One epoch, two absolute settings

The phase fit is centered on an internal reference epoch. Frequency error is evaluated at the separate intended application epoch. The old DDS must be the setting at that application epoch, not the setting at the start of the data or at the centered fit epoch.

The new external DDS is an **absolute setting**, not a number to add again. Its displayed increment is also reported in least-significant DDS digits. The new drift is the existing enabled drift plus a residual correction, with the sign and units handled explicitly. An existing disabled drift contributes zero regardless of a stale numeric value in its field.

The pp18/day input field and DDS-units/day input field are alternative representations of the same ramp. The program reports both new values and both increments. Do not apply both increments sequentially.

## Sign and precision

For the adopted timing definition, measured arrival(B) minus arrival(A) equals clock(A) minus clock(B). A direct MicroSem-to-GPS pair therefore uses the positive sign; a reverse pair uses the negative sign. This channel interpretation must be checked against the installed measurement wiring and logging convention.

Positive physical DDS units/day means the DDS number increases with time, which lowers output frequency [1]. The installed pp18 sign is separately selectable: positive raises output frequency, or positive raises DDS. The default is an explicit assumption, not a claim that Rev. J documents that sign.

The external DDS is rounded to seven decimal places. Both drift values are also shown without integer rounding and with nearest-integer alternatives. The integer alternatives are illustrative: verify what entry precision the installed masermon accepts. pp18 integer increments and DDS integer increments do not have the same effective frequency-drift resolution.

## Before entering any candidate

- **Reference**: confirm the GPS channel is healthy, not stepped or being steered unexpectedly.
- **Instrument**: confirm external synthesizer selection, current DDS, drift enable state and units.
- **Sign**: verify the instrument's installed pp18 convention; do not test it by an uncontrolled adjustment.
- **Time**: use a current, defensible application epoch and known setting at that epoch.
- **Fit**: inspect residuals, HMD changes, outages and window sensitivity.
- **Record**: preserve old settings, intended values, operator review and subsequent comparison data.

No checkbox turns this software into a certified controller. Values remain advisory even after the operator review fields are acknowledged.

## Stability analysis and Stable32

OADEV, MDEV, TDEV and OHDEV use phase samples in seconds and the NIST SP 1065 definitions [2]. The calculation supports raw phase, mean removal, linear phase removal and quadratic phase removal. Results are labeled with the chosen detrending; raw stability must not be confused with fitted-residual stability.

The code validates both adjacent intervals and cumulative deviation from a uniform grid. It never fills gaps or interpolates samples. The mean actual interval establishes the nominal sampling period after validating the record. The default relative tolerance is 1%; the interface allows up to 5%. This is an explicit approximate-uniformity tolerance, not proof that jitter has no effect at the shortest averaging times.

For the supplied month, a 2% tolerance admits an approximately 3 s maximum grid deviation on a 270 s cadence. The default 1% rejects the full record. An internal exclusion creates a missing interval and is not compressed away to make a false uniform series.

At least ten contributing terms are required for each plotted estimate. Overlapping terms are correlated; their count is not an equivalent number of independent degrees of freedom. No confidence bands or maser-only noise separation are claimed [2].




### Stable32 files

`raw.PHD` holds phase seconds and `raw.FRD` holds fractional frequency derived from successive phase samples. The frequency record has one fewer point. Fit-residual versions are written separately. The headers contain type, point count and tau; companion CSV/JSON files retain original timestamps, units, tolerance checks and export decisions [3].

Stable32 treats embedded zeros as gaps. Actual zeros are exported as `1e-99`, following the manual's recommendation; replacements are counted in the sidecar [3]. Do not use its gap-fill controls to disguise a nonuniform or incomplete input record.

# Output package and reproducibility

| Output | Meaning |
|---|---|
| `analysis.vszh5` | Native HDF5 document, typed datasets and Veusz plotting script. |
| `analysis.json` | Fit, coefficients, conditional covariance, settings, both drift units, provenance and warnings. |
| `phase_and_fit.csv` | All input timing samples, model, residual, selection and fit masks. |
| `telemetry_overlap.csv` | HMD rows overlapping the selected timing interval, if present. |
| `stability.csv` | Estimator, tau, deviation and overlapping-term count. |
| `raw.PHD / raw.FRD` | Optional uniform-record Stable32 phase and fractional frequency. |
| `fit_residual.*` | Separately identified residual stability inputs. |
| `review.md` | Human-readable advisory summary and reference table. |

The HDF5 document is written using the supplied writer without importing Veusz. All measured raw pair datasets and descriptive labels are included when the full collection is exported. Overview medians are separate datasets. The example's externally rendered PDF is a preview, not a dependency of the application.

Exports first build in a temporary adjacent folder and rename only after success. An existing destination is rejected. This protects earlier results on ordinary local filesystems; no distributed-filesystem transactional guarantee is claimed.

### Batch operation

```
python analyze_maser.py --fits-directory <FITS_DIRECTORY>
  --hmd <HMD_FILE> --start 2025-10-15 --end 2025-11-14
  --derive-dds --drift 975 --sampling-tolerance .02
  --stable32 --output <NEW_OUTPUT_DIRECTORY>
```

Join the lines for the shell. The end time is exclusive. Without DDS and drift inputs, the batch path exports measurement results without a correction candidate. `--derive-dds` only copies an HMD setting within 120 s of the historical application epoch; it does not retrieve present-day state.

### Writer revision

The attached writer was version 0.0.21. A targeted test demonstrated that `dataset_date()` accepted multidimensional input, creating invalid date shape or silently flattening it in resizable mode. Version 0.0.22 adds one-dimensional validation before creating the dataset group. Valid callers retain the same API. This is a scoped defect correction, not an exhaustive review of every writer feature.

## Historical validation results

The uploaded archive contains 109 FITS.GZ files and 40 measured channel pairs, 9,810 records per pair. The example selects October 15 through November 13, 2025, producing 9,600 direct MicroSem-to-CNS_GPS1 samples. The HMD record covers April 9, 2025, through March 25, 2026. The previously supplied July 2026 timing files do not overlap that HMD record.

At the last selected sample, November 13, 2025, 23:57:24.571 UTC, the quadratic fit gives frequency error $+5.83844 \times 10^{-15}$, residual drift $-660.833$ parts in $10^{18}$/day and phase-residual RMS 5.405 ns. These are computed results from the supplied files, not external instrument specifications.

**Review boundary.** The example assumes enabled drift $+975$ pp18/day and a positive pp18 sign that raises output frequency. The older screenshot supplies that example number, but does not establish its value in November. The following candidates are historical and unapproved.

| Quantity | Historical example |
|---|---|
| HMD DDS near the last timing sample | 5751.6772565 |
| Calculated DDS increment, last digits | +82.929597 |
| New rounded DDS | 5751.6772648 |
| New drift, pp18/day | +1635.833066 |
| Equivalent new drift, DDS units/day | -23.235467 |
| Nearest integer pp18/day alternative | +1636 |
| Nearest integer DDS units/day alternative | -23 |

### The dominant warning: fit-window sensitivity

| Window, d | Frequency error at common end | Drift, pp18/day | RMS, ns |
|---|---|---|---|
| 30 | $+5.83844 \times 10^{-15}$ | -660.833 | 5.405 |
| 21 | $+3.80025 \times 10^{-15}$ | -883.575 | 5.810 |
| 14 | $+1.39421 \times 10^{-14}$ | +407.199 | 5.922 |
| 7 | $-4.32655 \times 10^{-15}$ | -5454.180 | 6.970 |

The drift changes sign with the fitting interval. Conditional OLS errors are much smaller than this variation and must not be interpreted as a complete uncertainty budget. A longer/reference-checked operating-state review is needed before any live adjustment.

For the 30-day window, CNS2018 and CNS_GPS2 comparisons give $-688.966$ and $-652.571$ pp18/day. Similar values are useful but do not demonstrate independent references or isolate maser aging.

# Cross-checks, testing and limitations

## Franks/Veusz comparison

The supplied Veusz example was read as HDF5 data, without executing its embedded plotting script. Across 735 matching UTC-hour bins in the full archive span, the new derived MSM–GPS comparison differs by 1.284 ns RMS from the historical reduced values. MicroSem and CNS_GPS1 differ by 0.852 ns and 1.004 ns RMS. Median differences are approximately zero. The maximum absolute differences are 4, 2 and 4 ns respectively.

These small differences are consistent with nonidentical reduction windows and rounding, but that explanation is not a proof of identical processing. The direct raw pair remains the primary correction input. Invalid negative MJD legacy records and all-missing channels in the exemplar were not copied.

## Automated and rendering checks

The delivered suite has 36 passing tests on Python 3.12. Tests cover analytic phase coefficients, DDS direction, quantization, both drift representations, disabled drift, reversal, duplicates, gap/grid rejection, clipping, FITS metadata/padding, Franks sentinels, GUI controls/settings, writer behavior and export protection.

OADEV, MDEV, TDEV and OHDEV were compared with AllanTools for seeded synthetic phase data, including contributing-term counts. The 29-page historical `.vszh5` was opened and rendered with a separate Veusz installation. Its PDF has page bookmarks. GUI captures document the offscreen Qt tests.

## Not validated

- **Windows/bracca**: actual desktop operation and local staging were not performed.
- **Stable32 executable**: output structure was checked, but proprietary-application import remains a site test.
- **Installed masermon**: sign, range, precision, update cadence, persistence and firmware/version behavior remain site checks.
- **Physical system**: no live adjustment, calibration or post-adjustment monitoring was conducted.
- **Uncertainty**: no colored-noise confidence intervals, reference-error removal or independent maser-aging measurement is claimed.

## Repository provenance

The RF_Equipment_Comms stability-tool code [5] and FitsAutoPlot sources [6] were read for header, GUI and analysis conventions. The science modules in this package are new implementations using the cited definitions, not a wholesale copy of the signal-analyzer tool. Repository checkouts were read-only with respect to their remotes; no push was performed.

Reviewed commits:

- RF_Equipment_Comms: `9305e94e9ad85dbdb94a3765cf0989449dc3f345`.
- FitsAutoPlot: `36add38d973bc07e920ff70680a139241c421f7c`.

The product-page pointer is Microchip MHM-2010 [7].

# Appendix A: Implemented equations

### Time, phase and polynomial fit

Let $t$ be elapsed SI seconds, obtained by converting UTC timestamps through TAI, and $d$ elapsed days from fit epoch $t_0$. The MJD plotting conversion is only a civil-date coordinate; it is not used for elapsed-time derivatives.

$$t_i - t_0 = \mathrm{TAI}(\mathrm{UTC}_i) - \mathrm{TAI}(\mathrm{UTC}_0), \qquad d_i = (t_i - t_0)/86400, \tag{A1}$$

$$v_i = 86400(\mathrm{MJD}_i - 54832), \qquad h_i = \lfloor 24\,\mathrm{MJD}_i \rfloor. \tag{A2}$$

Here $v_i$ is the Veusz seconds coordinate relative to 2009-01-01 and $h_i$ indexes a UTC-hour reduction bucket. Within each nonempty bucket, time and phase are reduced by their medians:

$$(\widetilde{\mathrm{MJD}}_h, \widetilde{x}_h) = (\mathrm{median}_{i:h_i=h}\,\mathrm{MJD}_i,\ \mathrm{median}_{i:h_i=h}\,x_i)\,. \tag{A3}$$

With phase coefficients in nanoseconds and $d$ in days:

$$\widehat{x}(d) = 10^{-9}(c_0 + c_1 d + c_2 d^2)\ \mathrm{s}, \tag{A4}$$

$$y(d) = \frac{10^{-9}}{86400}(c_1 + 2c_2 d), \qquad r \equiv \frac{dy}{dd} = \frac{2c_2\,10^{-9}}{86400}\ \mathrm{day}^{-1}. \tag{A5}$$

For a linear fit, $c_2$ is not estimated; the drift result is withheld, not presented as evidence of zero physical drift.

Let $s = \max_i |d_i|$, $A_{ij} = (d_i/s)^j$, $b_i = 10^9 x_i$, and let $T = \mathrm{diag}(1, s^{-1}, s^{-2})$, truncated for a linear fit:

$$\widehat{\boldsymbol{\beta}} = \arg\min_{\boldsymbol{\beta}} \|A\boldsymbol{\beta} - \mathbf{b}\|_2^2, \qquad \mathbf{c} = T\widehat{\boldsymbol{\beta}}, \tag{A6}$$

$$\widehat{\sigma}^2 = \frac{\sum_i e_i^2}{N - p}, \qquad C_c = T\left[\widehat{\sigma}^2 (A^{\mathsf{T}}A)^{-1}\right] T^{\mathsf{T}}, \qquad e_i = b_i - (A\widehat{\boldsymbol{\beta}})_i. \tag{A7}$$

Here $p$ is coefficient count. The reported fit scatter and conditional uncertainties are:

$$\mathrm{RMS}_{\mathrm{ns}} = \sqrt{\frac{1}{N}\sum_i e_i^2}, \qquad \sigma_y = \frac{10^{-9}}{86400}\sqrt{\mathbf{g}^{\mathsf{T}} C_c \mathbf{g}}, \qquad \mathbf{g} = (0, 1, 2d)^{\mathsf{T}}, \tag{A8}$$

$$\sigma_r = \frac{2\,10^{-9}}{86400}\sqrt{(C_c)_{22}}. \tag{A9}$$

They assume IID residuals conditional on the chosen fit; they do not represent a full timing-standard uncertainty.

### Clipping, continuity and channel convention

Optional clipping uses median-centered residuals and a robust scale:

$$M = \text{median}(e), \quad S = 1.4826 \text{ median } |e - M|, \quad \text{retain } i \text{ if } |e_i - M| \le kS. \tag{A10}$$

There are at most eight clipping passes, followed by a final refit. $k = 0$ disables clipping. A fitting run is split at gaps exceeding five median timestamp intervals:

$$\Delta\text{MJD}_i > 5\,\text{median}(\Delta\text{MJD}). \tag{A11}$$

These are software heuristics, not manufacturer operating limits.

For ideal pulse arrival times and clock time errors relative to the same reference:

$$\Delta t_{A\to B} = t_{\text{arrival},B} - t_{\text{arrival},A} = x_A - x_B, \qquad x_{\text{maser}-\text{ref}} = \epsilon\,\Delta t, \quad \epsilon \in \{-1, +1\}. \tag{A12}$$

Fixed cable offsets affect the constant phase term; changing delays can affect slope and drift and are not removed automatically.

### DDS correction and both drift units

The manufacturer relationship is [1]:

$$f_o = \frac{f_m}{D}, \qquad D = 284.08 + aN, \qquad a = 2 \times 10^{-7}. \tag{A13}$$

For old DDS $N$, measured residual fractional frequency $y$, and least-significant DDS step $q = 10^{-7}$:

$$k_N = \frac{a}{D}, \qquad \frac{\delta f_o}{f_o} \simeq -k_N \delta N, \qquad g_{\text{digit}} = -k_N q, \tag{A14}$$

$$\Delta N = \frac{D}{a} y, \qquad N_{\text{new}} = \text{round}_q(N + \Delta N), \qquad \Delta n = \Delta N / q. \tag{A15}$$

Before rounding, the static relation exactly cancels the measured residual:

$$1 + y_{\text{new}} = (1 + y)\frac{D}{D_{\text{new}}}, \qquad D_{\text{new}} = 284.08 + aN_{\text{new}}. \tag{A16}$$

The quantization residual is computed from (A16) with Decimal arithmetic. The physical DDS rate $v$, in last digits/day, transforms as:

$$v_{\text{new}} = (1 + y)v_{\text{old}} + \frac{r}{k_N q}. \tag{A17}$$

Let $P$ be the displayed pp18/day value and $\eta = +1$ when positive $P$ raises output frequency, or $-1$ when it raises DDS:

$$v_{\text{old}} = -\frac{\eta P_{\text{old}} 10^{-18}}{k_N q}, \qquad P_{\text{new}} = -\eta\, v_{\text{new}}\, k_{N,\text{new}} q\, 10^{18}, \tag{A18}$$

$$P_{\text{new}} \simeq P_{\text{old}} - \frac{\eta r\, 10^{18}}{1 + y}, \qquad k_{N,\text{new}} = a / D_{\text{new}}. \tag{A19}$$

The code reports the rounded-DDS conversion in the both-units table. An inactive old drift uses $v_{\text{old}} = 0$. These equations describe the local rate at the application epoch, not undocumented firmware scheduling.

### Rounding, uniformity and Stable32 conversion

For nearest-integer drift alternatives:

$$\delta r_v = -(\text{round}\, v_{\text{new}} - v_{\text{new}}) k_{N,\text{new}} q, \tag{A20}$$

$$\delta r_P = \eta(\text{round}\, P_{\text{new}} - P_{\text{new}}) 10^{-18}. \tag{A21}$$

For a record of $N$ samples, interval tolerance $\rho$, and time coordinates from (A1), the validation is:

$$\tau_0 = \frac{t_{N-1} - t_0}{N-1}, \qquad \max_i \left| \frac{t_{i+1} - t_i}{\tau_0} - 1 \right| \le \rho, \tag{A22}$$

$$\max_i |t_i - t_0 - i\tau_0| \le \rho\tau_0, \qquad 0 < \rho \le 0.05. \tag{A23}$$

At least eight finite matching samples are required. No missing interval is compressed or interpolated. Stable32 fractional frequency and actual-zero encoding are:

$$\overline{y}_i = \frac{x_{i+1} - x_i}{\tau_0}, \qquad z_{\text{export}} = \begin{cases} 10^{-99}, & z = 0 \text{ (actual observation)}, \\ z, & z \neq 0. \end{cases} \tag{A24}$$

The zero convention follows Stable32's documented gap handling [3].

### Stability estimators

Let $\tau = m\tau_0$, with integer $m \ge 1$, and $x_i$ phase seconds. All sums are over valid overlapping terms of a validated complete record. The implementations follow NIST SP 1065 [2]:

$$\sigma_y^2(\tau) = \frac{\sum_{i=0}^{N-2m-1} (x_{i+2m} - 2x_{i+m} + x_i)^2}{2\tau^2(N-2m)}, \tag{A25}$$

$$\operatorname{mod}\sigma_y^2(\tau) = \frac{\displaystyle\sum_{j=0}^{N-3m} \left[ \sum_{i=j}^{j+m-1} (x_{i+2m} - 2x_{i+m} + x_i) \right]^2}{2m^2\tau^2(N-3m+1)}, \tag{A26}$$

$$\sigma_x(\tau) = \frac{\tau}{\sqrt{3}} \operatorname{mod}\sigma_y(\tau), \tag{A27}$$

$$\sigma_H^2(\tau) = \frac{\displaystyle\sum_{i=0}^{N-3m-1} (x_{i+3m} - 3x_{i+2m} + 3x_{i+m} - x_i)^2}{6\tau^2(N-3m)}. \tag{A28}$$

The displayed OADEV, MDEV and OHDEV are square roots of the respective variances. TDEV is (A27). Averaging factors are log-spaced integers, with duplicates removed and at least ten contributing terms per reported point. Optional detrending subtracts an OLS polynomial of order 0, 1 or 2 before applying these equations; raw mode removes only a constant for numerical convenience.

## What is new in release 0.2.0

- **T4 iMaser 3000**: maser selector; SETFSREL and DRIFTCOMP suggestions from the OEM relations [8], [9]; a T4 telemetry tab that reads recorder files and text logs and can poll the maser with read-only queries; a loopback emulator for tests and demonstrations.
- **Reference 1 PPS figures**: the operator's two figures (deltas with statistics; data and fit over residuals) in the GUI and as the first pages of every `analysis.vszh5`.
- **FITS and Franks together**: one selection may hold both; they remain separate series. Franks month-boundary repeats are resolved and reported; fits may bridge gaps.
- **Time axes**: every time-series page exists against MJD (UTC) and against the local date-time of the processing computer, with ISO 8601 tick labels [12] (writer 0.0.24).
- **GUI**: menu bar (File, View with Light/Dark theme, T4, Help).

## Reference 1 PPS figures

### Definitions

For the $N$ samples of a figure, $x_i = \mathrm{MJD}_i - \mathrm{MJD}_0$ (days, $\mathrm{MJD}_0$ the first sample shown) and $y_i$ is the 1 PPS time difference in ns. The deltas figure uses every sample of the selected interval; the fit figure uses the samples actually fitted (operator exclusions removed, drawn hollow and grey). The boxes report

$$\bar{y} = \frac{1}{N}\sum_i y_i, \qquad \mathrm{std} = \sqrt{\frac{1}{N}\sum_i (y_i - \bar{y})^2} \quad \text{(population value, as } \texttt{numpy.std}\text{)}.$$

### Straight-line fit and its standard errors

With $\bar{x}$, $S_{xx} = \sum (x_i - \bar{x})^2$ and $S_{xy} = \sum (x_i - \bar{x})(y_i - \bar{y})$, the least-squares line $y = a_1 x + a_0$ minimizes $\sum r_i^2$, $r_i = y_i - a_1 x_i - a_0$. Setting $\partial/\partial a_0 = \partial/\partial a_1 = 0$ gives the normal equations, whose solution is

$$a_1 = \frac{S_{xy}}{S_{xx}}, \qquad a_0 = \bar{y} - a_1 \bar{x}.$$

For independent residuals of variance $\sigma^2$, $\mathrm{Var}(a_1) = \sigma^2 / S_{xx}$; because $\mathrm{Cov}(\bar{y}, a_1) = 0$, $\mathrm{Var}(a_0) = \mathrm{Var}(\bar{y}) + \bar{x}^2 \,\mathrm{Var}(a_1) = \sigma^2 \left( \frac{1}{N} + \frac{\bar{x}^2}{S_{xx}} \right)$. With $s^2 = \sum r_i^2 / (N-2)$ in place of $\sigma^2$:

$$s(a_1) = \frac{s}{\sqrt{S_{xx}}}, \qquad s(a_0) = s\sqrt{\frac{1}{N} + \frac{\bar{x}^2}{S_{xx}}}, \qquad \mathrm{rate} = r = \frac{a_1\,[\mathrm{ns/day}] \times 10^{-9}}{86400\ \mathrm{s}}.$$

These are conditional IID errors; colored noise and GPS systematics are not included [2]. Check against the operator's figure: $N = 741$, $s = 3.49$ ns over 31 days gives $S_{xx} \approx 741 \times 31^2/12$, $s(a_1) \approx 0.014$ ns/day and $s(a_0) \approx 0.257$ ns, as printed.

### Synthesizer adjustment line

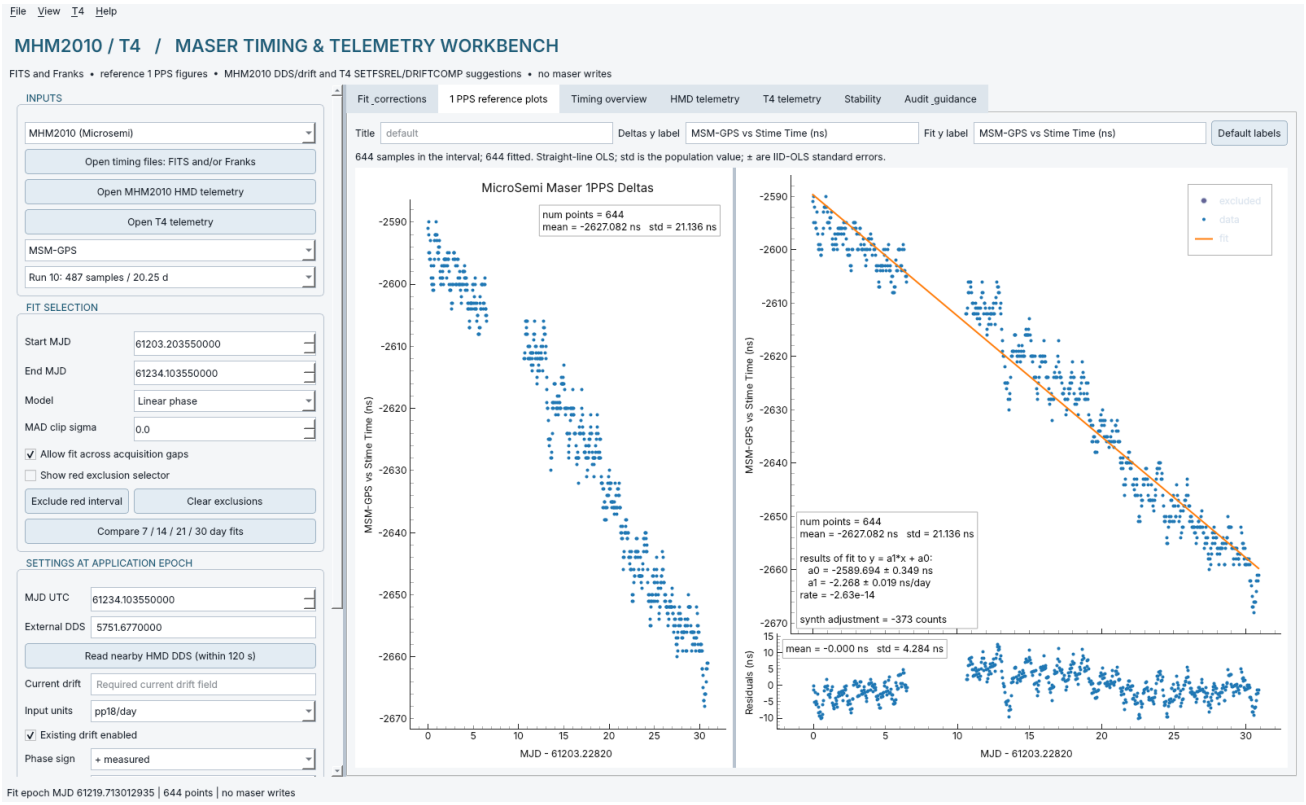
**MHM2010.** Appendix A gives the fractional change per DDS unit, $\partial y/\partial D = -2 \times 10^{-7}/(284.08 + 2 \times 10^{-7} D)$ [1]. One count is $10^{-7}$ DDS units, so cancelling $y = \sigma r$ ($\sigma = \pm 1$ the phase sign) requires

$$\Delta n = \frac{-\sigma r}{-2 \times 10^{-14}/(284.08 + 2 \times 10^{-7} D)} = \frac{\sigma r\,(284.08 + 2 \times 10^{-7} D)}{2 \times 10^{-14}}.$$

For $r = -3.15 \times 10^{-14}$ this is $-447$ counts, the value in the operator's figure; the full DDS calculation of Appendix A agrees to $10^{-9}$ relative (test suite).

**T4.** The line shows the SETFSREL entry and the synthesizer change of the next section for the same straight-line rate.

**Review boundary.** The figure's adjustment uses the straight-line mean rate over the plotted span, as the operator's figures do. The settings panel evaluates the selected model (linear or quadratic) at the application epoch. With a quadratic fit the two can differ; enter only the reviewed settings-panel values.




GUI tab "1 PPS reference plots" (Light theme) on the operator's Franks files, June–July 2026, MSM-GPS.

## FITS and Franks inputs together

- FITS series are raw channel pairs keyed "A -> B"; Franks series are hourly medians keyed by column (e.g. `MSM-GPS`). Keys cannot collide; nothing is merged or interpolated.
- **Month boundaries.** Consecutive `YYYY_MM.t00new` files repeat the first hours of the next month. Identical repeats are dropped. A disagreeing repeat (found in the supplied files at MJD 61222.01917, Site1Hz 0 vs 1) keeps the row from the file named for that row's month, and the conflict is listed in the audit tab.
- **Gaps.** The 0.1.0 rule refused a fit across an acquisition gap. "Allow fit across acquisition gaps" (default on for Franks data) fits through gaps and reports each one. The supplied 2026 MSM-GPS series has nine gaps of 5 to 99 h; inspect the residuals for a phase step.

# T4 iMaser 3000

## MONIT telemetry

The reply to `MONIT;` is `$` + 113 hexadecimal characters + CR LF (Type 1, Card FPGA T4S-724) or CR (Type 2, IO card LAN T4S-711) [8]. The channels are 32 of 12 bits, 8 of 8 bits and the main-PLL lock bit:

$$L_{\text{hex}} = 32 \cdot 3 + 8 \cdot 2 + 1 = 113, \qquad v_i = g_i\, c_i, \quad c_i = \texttt{int}(\text{field}_i, 16)\ \text{(unsigned)}.$$

Gains $g_i$ are Table 6 of [8], identical to the operator's repository table [10]; the $-15$ V rail has a negative gain. Channels 21, 37 and 39 are "Not used". Channel 40 (LOCK 100 MHz) is an 8-bit analog voltage; channel 41 is the lock bit (1 locked, 0 unlocked; any other code is flagged). Decoding the manual's example reply gives 4.765 V for channel 40; the operator's web-page screenshot shows 4.764 V.

**Inputs.** The recorder's `t4_raw_frames.csv` (re-decoded from the raw replies), its `t4_telemetry.csv` (raw codes re-scaled), or text logs with one `<time> $<frame>` per line. Times must be MJD or ISO 8601 with an offset.

**Live polling** (T4 tab) sends only `MONIT;`, `RDFS;`, `GETID;` and `RDREG=…;` over UDP port 14000 [9]. Each attempt uses a fresh socket with bounded retries; datagrams from other peers are ignored. Raw replies are written before decoding, into a folder stamped once at creation (`T4_telemetry_YYYYMMDDTHHMMSS±hhmm`). The preview buffer is bounded and can be flushed or offloaded to a pickle. A non-loopback address requires the approval box. "Use loopback emulator" serves the manual's example reply.

## Frequency setting: SETFSREL

The PLL forces the last IF to equal the synthesizer frequency, $\nu_{\text{synth}} = F_{\text{out}} - 284\,\nu_{\text{ocxo}}$, and the output is the 5 MHz OCXO, $\nu_0 = \nu_{\text{ocxo}} = (F_{\text{out}} - \nu_{\text{synth}})/284$ ([8], Eqs. 10–11). Differentiating at fixed $F_{\text{out}}$:

$$y_0 = \frac{\Delta\nu_0}{\nu_0} = \frac{-\Delta\nu_{\text{synth}}}{284 \times 5\ \text{MHz}} = \frac{-\Delta\nu_{\text{synth}}}{1.42 \times 10^9\ \text{Hz}} \qquad \text{(Eq. 12)}.$$

To cancel a measured offset $y$ the output must change by $\Delta y = -y$, so $\Delta\nu_{\text{synth}} = y \times 1.42 \times 10^9$ Hz. The synthesizer resolution is 39 bits of 5 MHz (Sec. 9.4), $\delta = 5\ \text{MHz}/2^{39} = 9.0949 \times 10^{-6}$ Hz ($6.405 \times 10^{-15}$ relative). The suggestion is quantized:

$$n = \text{round}\left(\frac{\Delta\nu_{\text{synth}}}{\delta}\right), \qquad \Delta y_{\text{realized}} = -\frac{n\,\delta}{1.42 \times 10^9\ \text{Hz}}, \qquad \texttt{SETFSREL} = \pm\Delta y_{\text{realized}}.$$

[9] defines SETFSREL as a relative step in s/s with a mandatory sign (range $\pm 29 \times 10^{-9}$), but not whether $+$ raises the output. The selector "SETFSREL sign" sets the convention, and the panel predicts the new `RDFS` value, $\nu_{\text{synth}} + n\delta$, which must lie within 405 708–405 795 Hz. **After entry, compare RDFS with the prediction**; a move in the opposite direction means the convention is reversed. RDFS values near 1.42 GHz (the web page's FREQ) are reduced by $1.42 \times 10^9$ Hz.

## Drift: DRIFTCOMP

With linear drift, $y(t) = y_0 + \dot{y}t$ and $x(t) = x_0 + y_0 t + \frac{1}{2}\dot{y}t^2$ ([8], Eqs. 22–23). The quadratic phase fit $x = c_0 + c_1 d + c_2 d^2$ (ns, days) gives $y = (c_1 + 2c_2 d) \times 10^{-9}/86400$ and a drift per day $\dot{y}_{\text{day}} = 2c_2 \times 10^{-9}/86400$. "A positive sign will generate a positive slope compensation" [9], so

$$\texttt{DRIFTCOMP}_{\text{new}} = \texttt{DRIFTCOMP}_{\text{now}} - \sigma\,\dot{y}_{\text{day}}, \qquad 10^{-17} < |\texttt{DRIFTCOMP}| < 8 \times 10^{-13},$$

followed by REBOOT of the LAN module and `RDREG=DRIFTCOMP;` to verify. Neither command removes the 1 PPS time offset (ADJ1PPS/SYNCPPS).

| Demonstration (MSM-GPS used as a stand-in, *not* T4 data) | Value |
|---|---|
| Fit: last 30.9 d of 2026 Franks MSM-GPS, quadratic, gap bridged | 644 samples |
| $y$ at MJD 61234.10355 | $-3.7365 \times 10^{-14}$ |
| $\Delta\nu_{\mathrm{synth}}$, steps | $-54.570$ $\mu$Hz, $-6$ |
| Suggested entry ("+ raises output") | `SETFSREL=+3.843E-14;` |
| RDFS before → predicted after | 405750.927598720 → 405750.927544150 Hz |
| Drift entry (current 0) | `DRIFTCOMP=+7.11e-16;` |

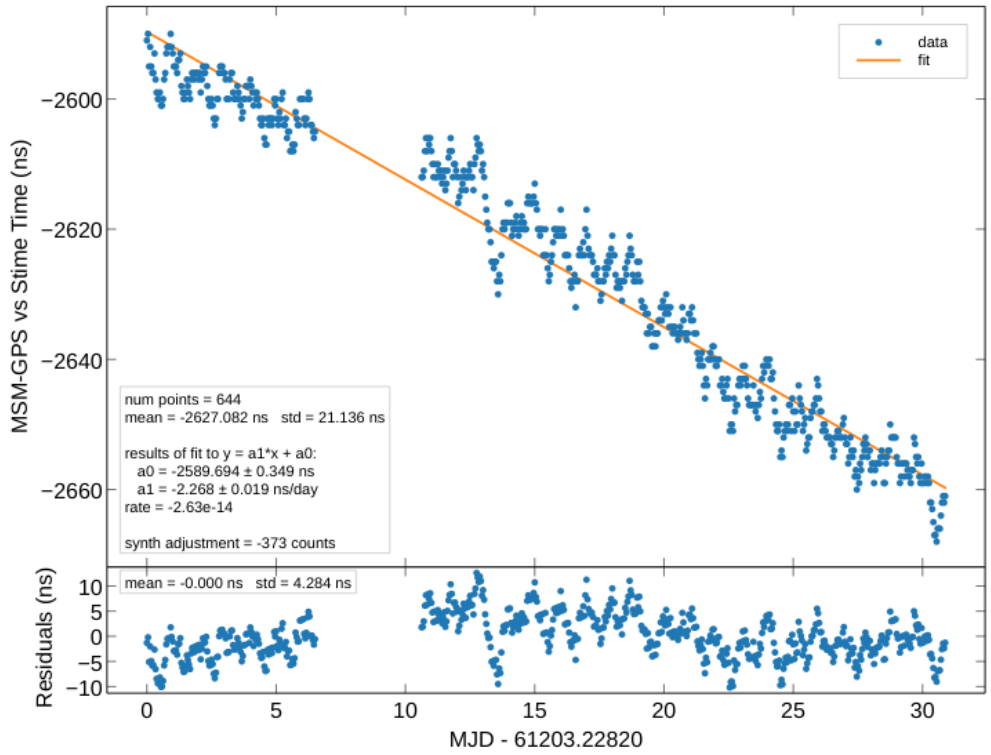
**Review boundary.** No FITS pair or Franks column is identified as the T4 yet. Designate the channel in the GUI ("Selected channel is the T4 comparison") or with `--maser T4 --channel`. T4 settings stay blocked until the channel is designated.

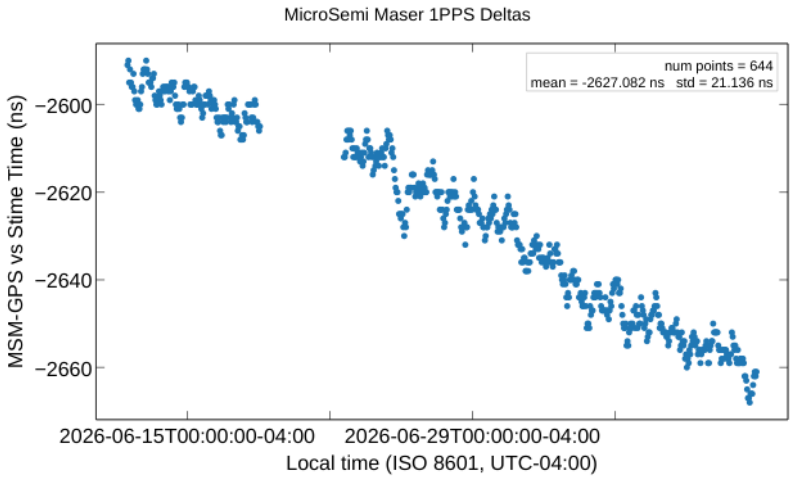
## MJD and local ISO 8601 time axes

Veusz stores dates as naive seconds from 2009-01-01; a local wall-clock axis is therefore

$$V_i = t_i^{\mathrm{Unix}} - 1\,230\,768\,000\ \mathrm{s} + \Delta_{\mathrm{local}}(t_i), \qquad t^{\mathrm{Unix}} = (\mathrm{MJD} - 40587) \times 86400\ \mathrm{s},$$

where $\Delta_{\mathrm{local}}$ is the processing computer's UTC offset, evaluated per sample (daylight saving aware). The tick format is `%VDY-%VDm-%VDdT%VDH:%VDM:%VDS` followed by the offset, e.g. `2026-11-03T06:31:15-05:00`. Veusz formats are static, so when the data cross a DST change the ticks omit the offset and the axis label lists each offset and when it begins. Each time-series page is written twice (`…_mjd`, `…_local`); the reference figures keep the operator's MJD − $\mathrm{MJD}_0$ axis and gain `pps_deltas_local` and `pps_fit_local`. Writer helpers: `mjd_to_epoch`, `local_time_axis`, `iso_local_strings`, `time_pair_pages`.






Rendered by Veusz 3.6.2 from `analysis.vszh5`: fit page against MJD − $\mathrm{MJD}_0$, and the deltas twin against local ISO time (EDT in June).

## Validation of release 0.2.0

| Check | Result |
|---|---|
| Test suite (Python 3.12.15, PySide6 6.12.0) | 79 passed: 36 retained from 0.1.0, 29 T4, 10 reference/Franks/time/export, 4 GUI |
| Manual example MONIT replies (Type 1 and 2) | Decoded; channel 40 4.765 V vs 4.764 V on the web page |
| Franks 2026, MSM-GPS, last 30.9 d, linear | $N = 644$, $a_1 = -2.268 \pm 0.019$ ns/day, rate $-2.63 \times 10^{-14}$, $-373$ counts (DDS calculation: $-372.86$) |
| Writer 0.0.23 master | Rebuilt byte-exact (SHA-256 `90833d76…eaac6`); 0.0.24 adds helpers only |
| Veusz 3.6.2 renders | 25 pages incl. reference, overview and local ISO pages |
| UDP fault cases (emulator) | timeout, exhaustion, wrong peer, duplicate, garbage, truncation, port in use, cancellation, CR-only replies |

**Not validated:** a real T4 (no GBO captures yet), the SETFSREL sign, Windows, Stable32 import, and Veusz 4.2.1 for the new pages.

## Release checklist

- **Code**: Python 3.12 syntax, headers, revision history and Spyder cells.
- **GUI**: qtpy/PySide6, pyqtgraph, fitting, overview, HMD, stability and audit.
- **Science**: explicit phase sign, epoch, both drift units, no housekeeping-only calibration.
- **Outputs**: native HDF5, raw and reduced datasets, Stable32 and provenance.
- **Documentation**: editable LuaLaTeX, references, equations and PDF bookmarks.
- **Limits**: no live writes; Windows, masermon and Stable32 site tests remain.

Release 0.2.0 adds the T4 iMaser 3000 (read-only telemetry and advisory settings). Site tests on the real T4 (captured MONIT replies, RDFS before and after a SETFSREL entry) and on Windows remain open; see Section "Validation of release 0.2.0".

## IEEE reference table

Online references were accessed November 3, 2026. The source URLs are retained in full; absent DOIs or unverified publication details are not invented.


| Ref. | Full bibliographic entry | DOI or URL |
|---|---|---|
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| [2] | W. J. Riley, *Handbook of Frequency Stability Analysis*, NIST Special Publication 1065, Jul. 2008. | https://tf.nist.gov/general/pdf/2220.pdf |
| [3] | W. J. Riley, *Stable32 User Manual*, ver. 1.54, Sep. 7, 2008. | http://www.stable32.com/Manual154.pdf |
| [4] | F. Ghigo, "Clock Monitoring at Green Bank," National Radio Astronomy Observatory, web page; revision information varies within the page. | https://www.gb.nrao.edu/~fghigo/timer/time_logging.html |
| [5] | W. W. Wallace, "RF_Equipment_Comms: sa_stability_tool," GitHub repository, reviewed commit `9305e94e9ad85dbdb94a3765cf0989449dc3f345`, accessed Nov. 3, 2026. | https://github.com/Braveheart53/RF_Equipment_Comms/tree/main/sa_stability_tool |
| [6] | W. W. Wallace, "FitsAutoPlot," GitHub repository, reviewed commit `36add38d973bc07e920ff70680a139241c421f7c`, accessed Nov. 3, 2026. | https://github.com/Braveheart53/FitsAutoPlot |
| [7] | Microchip Technology, "MHM-2010," product web page, publication date not stated, accessed Nov. 3, 2026. | https://www.microchip.com/en-us/product/mhm-2010 |
| [8] | T4Science SA, *iMaser 3000 Installation, Operation and Maintenance User Manual*, doc. T4S-MAN-0012, issue 1.27, May 2, 2022, Secs. 9.4, 9.6, 9.12, 9.13 (Table 6), pp. 35–49. Supplied by the operator. | Not publicly hosted; operator copy (PDF), Nov. 3, 2026. |
| [9] | T4Science SA, *T4S-Monitoring Network Software User Manual*, doc. T4S-MAN-0015, ver. 1.17, Jul. 19, 2022, Secs. 1.1.1–1.1.3. | Not publicly hosted; operator copy (PDF), Nov. 3, 2026. |
| [10] | W. W. Wallace, "GBO_Maser_Control: T4_Maser_Comms.py, T4_channel_names.csv, T4_fromOEM.py," GitHub repository, reviewed commit `c72244d`, accessed Nov. 3, 2026. | https://github.com/Braveheart53/GBO_Maser_Control |
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| [12] | ISO, *Date and time — Representations for information interchange — Part 1: Basic rules*, ISO 8601-1:2019. | https://www.iso.org/standard/70907.html |